Semiparametric Efficiency Bounds for Networks

Seung Hyun Kim

Princeton University

October 13, 2026

Contents

Introduction

Multi-Rate Semiparametric Efficiency Theory

Efficient Inference on the Mean in Network Models

- Efficiency Gains for the Simple Model
- A Non-Degenerate CID Model
- A Non-Degenerate CID Model with Additive Link Function

Conclusion

Appendix

- Proofs of Main Results
- Efficient Inference in Network Models with Exogenous Variables

Network Data in Economics

- Network/dyadic data are prevalent in many areas of economics
  - Models of peer effects (Manski (1993); Bramoullé, Djebbari, and Fortin (2009); Jochmans (2023))
  - Dyadic regressions in areas such as international trade (Tinbergen (1963); Silva and Tenreyro (2006); Fafchamps and Gubert (2007))
  - Models of network formation (Graham (2017) ; Miyauchi (2016); Gao, Li, and Xu (2023))
- Despite prevalence, little is known about semiparametric efficiency for network data—previous attempts have assumed Gaussianity and an additive error components structure or proved efficiency only for models belonging to the exponential family class
- Lack of results can be attributed to unique dependence structure that induces tangent directions that **converge at different rates**

The Multi-Rate Problem in a Simple Model

- Given dyadic data $\{Y_{ij}\}_{i,j\in\mathbb{N}_+}$ and unit-specific characteristics $\{X_i\}_{i\in\mathbb{N}_+}$ suppose that the model is given as

$$Y_{ij} = X_i + X_j + \varepsilon_{ij}, \quad \{\varepsilon_{ij}\}_{i,j\in\mathbb{N}_+} \perp \{X_i\}_{i\in\mathbb{N}_+}$$

- n-sample score functions for one-dimensionals submodels in this case are of the form

$$\mathcal{S}_n = \sum_{i<j}^{n} q_\varepsilon(Y_{ij} - X_i - X_j) + \sum_{i=1}^{n} q_X(X_i)$$

for the edge-level score $q_\varepsilon(\cdot)$ (score of the shocks ε_{ij}) and the node-level score $q_X(\cdot)$ (score of the characteristics X_i) satisfying the zero-mean restrictions

$$\mathbb{E}_{P_0}\left[q_\varepsilon(\varepsilon_{ij})\right] = \mathbb{E}_{P_0}\left[q_X(X_i)\right] = 0$$

The Multi-Rate Problem in a Simple Model

- Consider two cases:
  i) **No Edge-Level Component**
  If $q_\varepsilon(\cdot) = 0$ so that the one-dimensional submodel corresponds to a pure node level perturbation, then $\mathcal{S}_n$ is the sum of n entries—**Converges at rate** $\sqrt{n}$

  ii) **No Node-Level Component**
  If $q_X(\cdot) = 0$ so that the one-dimensional submodel corresponds to a pure edge level perturbation, then $\mathcal{S}_n$ is the sum of $\binom{n}{2}$ entries—**Converges at rate** n

- In the usual iid case all tangent directions converge at the same $\sqrt{n}$ rate, so that any efficient estimator also converges at the rate $\sqrt{n}$: this reduces the question of efficiency to finding the smallest asymptotic variance

- Even in this kind of simple model the rate of the efficient estimator is a priori unknown

An Efficiency Framework for Multi-Rate Models

- I propose a new framework to deal with models where tangent directions converge at potentially different rates

- Approach the problem in two steps:
  1. For each possible rate of convergence n^{γ} collect tangent directions that converge at the n^{γ} rate into a "stratum" $\dot{\mathcal{P}}_P(\gamma)$—and then find the efficient rate of convergence n^{γ^*}
  2. Within the stratum $\dot{\mathcal{P}}_P(\gamma^*)$ apply usual semiparametric efficiency theory to find the efficiency bound

- Main results:
  Additional model restrictions may change not only the efficiency bound but the efficient rate;

  For general non-degenerate conditionally independent dyad (CID) models the sample mean is efficient for the dyad mean;

  Knowledge of the link function does not improve on efficiency—the sample mean is adaptive

Comparison of Classical v.s. Multi-Rate SP Theory

Classical SP Theory Choose a common convergence rate (e.g. $\sqrt{n}$) $\rightarrow$ Construct tangent space (often Hilbert) $\rightarrow$ Find efficient score via projection $\rightarrow$ Obtain efficiency bound

Multi-Rate SP Theory Define tangent set as collection of scores and their convergence rates $\rightarrow$ Find the efficient rate $\rightarrow$ Apply classical theory to "stratum" of tangent set collecting scores with convergence rate equal to the efficient rate $\rightarrow$ Obtain efficiency bound

The multi-rate framework is necessary when admissible tangent directions operate at different convergence rates, so the efficient rate cannot be fixed a priori

Contents

Introduction

Multi-Rate Semiparametric Efficiency Theory

Efficient Inference on the Mean in Network Models

- Efficiency Gains for the Simple Model
- A Non-Degenerate CID Model
- A Non-Degenerate CID Model with Additive Link Function

Conclusion

Appendix

- Proofs of Main Results
- Efficient Inference in Network Models with Exogenous Variables

Setup and Notation

- Data $\{Y_i\}_{i \in N_+}$, true DGP P_0 and consider one-dimensional submodels $\{P_r \mid r \geq 0\}$
- One-dimensional submodel $\{P_r \mid r \geq 0\}$ is said to be admissible if it satisfies model restrictions—collect these in set $\mathcal{P}_{sp}$
- Any n^γ for some $\gamma > 0$ is called a **convergence rate**
- Tangent sets now record two components: the (scaled) n-sample score sequence and a corresponding convergence rate
- We are interested in the k-dimensional parameter $\beta(P)$—$\beta_0 = \beta(P_0)$ is its true value

Tangent Set

Definition 1 (Maximal Tangent Set)

The maximal tangent set $\dot{\mathcal{P}}_{max}$ collects pairs $\{n^{\gamma}, S_n\}_{n \in N_+}$ such that:

- There exists some $\mathbb{I}(S) > 0$ such that

$$S_n \overset{P_0}{\Rightarrow} \mathcal{N}\left[0, \mathbb{I}(S)\right], \quad \mathbb{E}_{P_0}\left[S_n^2\right] \to \mathbb{I}(S)$$

- There exists some one-dimensional submodel $\{P_r \mid r \geq 0\}$ such that

$$\log \frac{dP_{n,n^{-\gamma}}}{dP_{n,0}}(\mathbf{Y}_n) = S_n - \frac{1}{2}\mathbb{I}(S) + o_{P_0}(1)$$

where $P_{n,r}$ is the n-sample law under P_r evaluated at $r \geq 0$

The semiparametrically relevant tangent set $\dot{\mathcal{P}}_P$ is a subset of $\dot{\mathcal{P}}_{max}$ collecting rate/score pairs corresponding to admissible submodels

Running Example: The Simple Additive Model

- Consider the aforementioned simple model:

$$Y_{ij} = X_i + X_j + \varepsilon_{ij}, \quad \{\varepsilon_{ij}\}_{i,j\in\mathbb{N}_+} \perp \{X_i\}_{i\in\mathbb{N}_+}$$

with $\mathbb{E}_{P_0}[X_i] = \mu_X, \mathrm{Var}_{P_0}[X_i] = \sigma_X^2 > 0$

- The n-sample likelihood corresponding to the submodel P_r is given as

$$p(\mathbf{Y}_n, \mathbf{X}_n; r) = p(\mathbf{Y}_n \mid \mathbf{X}_n; r)p(\mathbf{X}_n; r) = \left\{\prod_{i<j}^{n} p(Y_{ij} \mid X_i, X_j; r)\right\}\left\{\prod_{i=1}^{n} p(X_i; r)\right\}$$

- By the independence between X_i, X_j and ε_{ij} the conditional density $p(Y_{ij} \mid X_i, X_j; r)$ is the density of ε_{ij} evaluated at $Y_{ij} - X_i - X_j$

- Taking logs on both sides and differentiating shows us that there exist scores $q_\varepsilon(\cdot), q_X(\cdot)$ such that

$$\partial_r \log p(\mathbf{Y}_n, \mathbf{X}_n; r)|_{r=0} = \sum_{i<j}^{n} q_\varepsilon(Y_{ij} - X_i - X_j) + \sum_{i=1}^{n} q_X(X_i)$$

Running Example: The Simple Additive Model

- The semiparametrically relevant tangent set $\dot{\mathcal{P}}_P$ for the simple model is a subset of the maximal tangent set $\dot{\mathcal{P}}_{max}$ where the pair $\{n^\gamma, S_n\}_{n \in N_+}$ satisfies

$$S_n = \frac{1}{n^\gamma} \left\{ \sum_{i<j}^{n} q_\varepsilon(Y_{ij} - X_i - X_j) + \sum_{i=1}^{n} q_X(X_i) \right\}$$

  for square integrable functions $q_\varepsilon(\cdot), q_X(\cdot)$ such that $\mathbb{E}_{P_0}\left[q_\varepsilon(\varepsilon_{ij})\right] = \mathbb{E}_{P_0}\left[q_X(X_i)\right] = 0$

- We can immediately construct score sequences converging at different rates:

  1) **Convergence Rate** $\sqrt{n}$
     If $q_\varepsilon(\cdot) = 0$ and $\mathbb{E}_{P_0}\left[q_X(X_i)^2\right] > 0$ then $\{\sqrt{n}, S_n\}_{n \in N_+} \in \dot{\mathcal{P}}_P$

  2) **Convergence Rate** n
     If $q_X(\cdot) = 0$ and $\mathbb{E}_{P_0}\left[q_\varepsilon(\varepsilon_{ij})^2\right] > 0$ then $\{n, S_n\}_{n \in N_+} \in \dot{\mathcal{P}}_P$

Differentiability of Parameter Mapping

Assume that the parameter mapping $\beta(\cdot)$ is differentiable with derivative that depends only on the rate/score pair of a one-dimensional submodel:

Definition 2 (Directional Differentiability)

There exists a mapping $\mathcal{G} : \dot{\mathcal{P}}_P \to \mathbb{R}^k$ such that, for any $\{P_r \mid r \geq 0\} \in \mathcal{P}_{sp}$ and corresponding rate/score pair $\{n^\gamma, S_n\}_{n \in N_+} \in \dot{\mathcal{P}}_P$,

$$\lim_{r \to 0} \frac{\beta(P_r) - \beta(P_0)}{r} = \mathcal{G}(\{n^\gamma, S_n\}_{n \in N_+}).$$

Running Example: The Simple Additive Model

- The parameter of interest is $\beta(P) = \mathbb{E}_P\left[Y_{ij}\right]$ with $\beta_0 = \beta(P_0)$
- For any submodel P_r with corresponding rate/score pair $\{n^\gamma, S_n\}_n \in \dot{\mathcal{P}}_P$,

$$\frac{\partial}{\partial r}\beta(P_r)|_{r=0} = n^\gamma \mathbb{E}_{P_0}\left[\bar{Y}_n S_n\right] = n^\gamma \mathbb{E}_{P_0}\left[(\bar{Y}_n - \beta_0)S_n\right]$$

- Since the variance of X_i is positive under P_0, the CLT reveals that

$$\begin{aligned}\sqrt{n}(\bar{Y}_n - \beta_0) &= \frac{2}{\sqrt{n}}\sum_{i=1}^{n}(X_i - \mu_X) + \frac{2}{\sqrt{n}(n-1)}\sum_{i<j}^{n}(\varepsilon_{ij} - \mu_\varepsilon) \\ &\overset{P_0}{\Rightarrow} \mathcal{N}\left[0, 4\sigma_X^2\right]\end{aligned}$$

- $\mathcal{G}(\cdot)$ becomes

$$\mathcal{G}(\{n^\gamma, S_n\}_{n\in N_+}) = n^{\gamma-\frac{1}{2}} \underbrace{\mathbb{E}_{P_0}\left[\sqrt{n}(\bar{Y}_n - \beta_0)S_n\right]}_{O(1)} \quad \forall n$$

Efficiency Bound as Largest CRLB

- Letting $\beta(\cdot)$ be scalar valued, a straightforward generalization of the classical efficiency bound would be to define the bound as

$$\sup_{\{n^{\gamma}, S_n\}_{n \in N_+} \in \dot{\mathcal{P}}_P} \frac{\mathcal{G}(\{n^{\gamma}, S_n\}_{n \in N_+})^2}{\mathbb{I}(S)}$$

- But for efficiency the rate takes precedence over the asymptotic variance...and so we want to define the efficiency bound as

$$\sup_{\{n^{\gamma^*}, S_n\}_{n \in N_+} \in \dot{\mathcal{P}}_P} \frac{\mathcal{G}(\{n^{\gamma^*}, S_n\}_{n \in N_+})^2}{\mathbb{I}(S)}$$

  for a convergence rate n^{γ^*} that is efficient in some sense

Stratifying the Tangent Set

Definition 3 (Strata of Score Directions)

Given some $\gamma > 0$, we define

$$\dot{\mathcal{P}}_P(\gamma) = \left\{ \{S_n\}_{n \in N_+} \mid \{n^\gamma, S_n\}_{n \in N_+} \in \dot{\mathcal{P}}_P \right\}$$

Definition 4 (Stratum-Specific Modulus of Derivative)

For each $\gamma > 0$, we define the modulus of the derivative mapping $\mathcal{G}(\cdot)$ at the stratum $\dot{\mathcal{P}}_P(\gamma)$ as

$$V(\gamma) = \sup_{\{S_n\}_{n \in N_+} \in \dot{\mathcal{P}}_P(\gamma)} \left| \mathcal{G}(\{n^\gamma, S_n\}_{n \in N_+}) \right|$$

where we define the supremum of the empty set to be equal to 0.

If $V(\gamma) = 0$ then no tangent direction that converges at the rate n^γ contains information about β because the rate induces tangent directions that are too "flat" in some sense—so we want to focus our attention on rates that induce enough curvature

The Leading First-Order Rate

- If $V(\gamma) = 0$ we say that the stratum $\dot{\mathcal{P}}_P(\gamma)$ is first-order noninformative
- The leading first-order rate is defined as the slowest convergence rate for which any slower strata are first-order noninformative

Definition 5 (Leading First-Order Rate)

Define

$$\gamma^* = \inf \left\{ \gamma > 0 \mid V(\gamma) > 0 \right\}$$

with the convention that the infimum of the empty set is $+\infty$. If $\gamma^* \in (0, +\infty)$ then n^{γ^*} is called the leading first-order rate.

Running Example: The Simple Additive Model

Derivative Mapping: $\mathcal{G}(\{n^\gamma, S_n\}_{n\in N_+}) = n^{\gamma-\frac{1}{2}} \underbrace{\mathbb{E}_{P_0}\left[\sqrt{n}(\bar{Y}_n - \beta_0)S_n\right]}_{O(1)} \quad \forall n$

- For any $\gamma < \frac{1}{2}$, we can see that $\mathcal{G}(\{n^\gamma, S_n\}_{n\in N_+}) = 0$ because it is the product of $o(1)$ and $O(1)$ terms—thus any stratum $\dot{\mathcal{P}}_P(\gamma)$ with $\gamma < \frac{1}{2}$ is first-order non-informative (equivalently, $V(\gamma) = 0$)
- The rate/score pair $\{n^{1/2}, S_n\}_{n\in N_+}$ defined as

$$S_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} (X_i - \mu_X)$$

  is contained in $\dot{\mathcal{P}}_P$ and $\mathcal{G}(\{n^{1/2}, S_n\}_{n\in N_+}) \neq 0$, so the stratum $\dot{\mathcal{P}}_P(1/2)$ is first-order informative
- Therefore, $\gamma^* = \frac{1}{2}$ and $n^{1/2}$ is the leading first-order rate—we restrict our attention to tangent directions that converge at least as fast as $n^{1/2}$

Regular Estimators and the Efficient Rate

A regular estimator is an estimator whose limiting distribution is invariant against rate/score pairs that converge at least as fast as the leading first-order rate

Definition 6 (Regularity)

Let n^{γ^*} be the leading first-order rate. An estimator $\mathcal{B}_n(\mathbf{Y}_n)$ of β is regular if there exists a distribution $\mathcal{L}$ such that

$$n^{\kappa}\left\{\mathcal{B}_n(\mathbf{Y}_n) - \beta(P_{n^{-\gamma}})\right\} \overset{P_{n^{-\gamma}}}{\Rightarrow} \mathcal{L}$$

for any $\{P_r \mid r \geq 0\} \in \mathcal{P}_{sp}$ with corresponding rate/score pair $\{n^{\gamma}, S_n\}_{n \in N_+} \in \dot{\mathcal{P}}_P$ satisfying $\gamma \geq \gamma^*$.

n^{γ^*} is the leading first-order rate;
n^{κ} is the convergence rate of the estimator;
n^{γ} is the convergence rate of a local alternative;
We define regularity for local alternatives satisfying $\gamma \geq \gamma^*$

Regular Estimators and the Efficient Rate

- The efficient rate is then the fastest possible convergence rate for a regular estimator whose stratum is first-order informative

Definition 7 (The Efficient Rate of Convergence)

n^{γ} is said to be the efficient rate of convergence if

- There exists some score sequence $\{S_n\}_{n \in N_+}$ in the stratum $\dot{\mathcal{P}}_P(\gamma)$ such that $\mathcal{G}(\{n^{\gamma}, S_n\}_{n \in N_+}) \neq O_{k \times 1}$.
- For any regular estimator $\mathcal{B}_n(\mathbf{Y}_n)$ of β that converges at the rate n^{κ}, we have $\kappa \leq \gamma$

Relationship between Leading First-order and Efficient Rates

Lemma 1

Let n^{γ^*} be the leading first-order rate, and suppose $\mathcal{B}_n(\mathbf{Y}_n)$ is a regular estimator for β with rate of convergence n^{κ}. If $\kappa > \gamma^*$, then $\dot{\mathcal{P}}_P(\gamma^*)$ is first-order noninformative, that is,

$$\mathcal{G}(\{n^{\gamma^*}, S_n\}_{n \in N_+}) = O_{k \times 1} \quad \forall \{S_n\}_{n \in N_+} \in \dot{\mathcal{P}}_P(\gamma^*).$$

Proof Sketch

Theorem 1 (Sufficient Conditions for Efficiency)

Let n^{γ^*} be the leading first-order rate. If the stratum $\dot{\mathcal{P}}_P(\gamma^*)$ is first-order informative, then n^{γ^*} is the efficient rate of convergence.

If $\dot{\mathcal{P}}_P(\gamma^*)$ is first order informative, then the efficiency bound is going to be defined as the maximum CRLB at the stratum $\dot{\mathcal{P}}_P(\gamma^*)$: in the scalar case,

$$V_0 = \sup_{\{S_n\}_{n \in N_+}} \frac{\mathcal{G}(\{n^{\gamma^*}, S_n\}_{n \in N_+})^2}{\mathbb{I}(S)}$$

The Efficiency Bound

Convolution theorem then holds if we can find an efficient score sequence Ψ_n for the derivative mapping $\mathcal{G}(\cdot)$ in the **efficient stratum** $\dot{\mathcal{P}}_P(\gamma^*)$

Theorem 2 (Multi-Rate Convolution Theorem)

Let n^{γ^*} be the leading first-order rate with a first order informative stratum $\dot{\mathcal{P}}_P(\gamma^*)$, so that n^{γ^*} is also the efficient rate of convergence. Suppose that there exists a sequence $\{\Psi_n\}_{n\in N_+}$ of k-dimensional random vectors with limiting positive definite covariance matrix V_0 such that, for any non-zero $h \in \mathbb{R}^k$,

i) ***Tangency Condition.*** $\{h'\Psi_n\}_{n\in N_+} \in \dot{\mathcal{P}}_P(\gamma^*)$.

ii) ***Local Efficiency Condition.*** $\mathcal{G}(\{n^{\gamma^*}, h'\Psi_n\}_{n\in N_+}) = V_0 h$

Then, for any regular estimator $\mathcal{B}_n(\mathbf{Y}_n)$ of β that converges at the rate n^{γ^*} with limiting distribution $\mathcal{L}$, there exists a probability measure M such that $\mathcal{L} = \mathcal{N}[O_{k\times 1}, V_0] * M$.

Proof Sketch

Running Example: The Simple Additive Model

- The stratum $\dot{\mathcal{P}}_P(1/2)$ is first-order informative, so by theorem 1 $\sqrt{n}$ is the efficient convergence rate
- Defining $\Psi_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} 2(X_i - \mu_X)$, $\{\Psi_n\}_n$ satisfies the tangency and local efficiency conditions:
  1) For any non-zero $a \in \mathbb{R}$, letting $q_\varepsilon(\cdot) = 0$ and $q_X(x) = 2a(x - \mu_X)$ shows us that $\{a\Psi_n\}_{n \in N_+} \in \dot{\mathcal{P}}_P(1/2)$
  2) Since $\sqrt{n}(\bar{Y}_n - \beta_0) = \Psi_n + o_{L^2}(1)$,

     $$\mathcal{G}(\{n^{1/2}, S_n\}_{n \in N_+}) = \lim_{n \to \infty} \mathbb{E}_{P_0}\left[\sqrt{n}(\bar{Y}_n - \beta_0) S_n\right] = \lim_{n \to \infty} \mathbb{E}_{P_0}\left[\Psi_n S_n\right]$$

     for any $\{S_n\}_{n \in N_+} \in \dot{\mathcal{P}}_P(1/2)$
- Therefore, the variance $4\sigma_X^2$ of Ψ_n is the efficiency bound by theorem 2

The Problem of Attainment

- In classical SP theory, efficient score immediately defines efficient estimator: efficiency condition holds for any tangent direction + joint normality holds by Lindeberg-Levy CLT
  $\Rightarrow$ Estimator that is asymptotically linear in efficient score is regular
- Efficiency condition in theorem 2 holds only for tangent directions that are linear combinations of the efficient score Ψ_n (hence "local") + joint normality for Ψ_n and some tangent direction S_n may not hold
  $\Rightarrow$ Regularity of efficient score does not follow immediately
- In the multi-rate setting, the problems of finding the efficiency bound and attainment of the bound are separated

Theorem 3 (Sufficient Conditions for Attainment)

Let n^{γ^*} be the leading first order rate and suppose $\dot{\mathcal{P}}_P(\gamma^*)$ is first-order informative, so that n^{γ^*} is the efficient rate of convergence as well. Let $\{\Psi_n\}_{n\in N_+}$ be a sequence of k-dimensional random vectors satisfying the tangency and local efficiency conditions.
Suppose that, for any $\{n^\gamma, S_n\}_{n\in N_+} \in \dot{\mathcal{P}}_P$ such that $\gamma \geq \gamma^*$,

1) ***Joint Normality***
   (Ψ_n, S_n) converge weakly under P_0 to a normal distribution as $n \to \infty$.
2) ***Global Efficiency***
   $$\left|\mathbb{E}_{P_0}[\Psi_n S_n] - n^{\gamma^* - \gamma}\mathcal{G}(\{n^\gamma, S_n\}_{n\in N_+})\right| = o(1) \text{ as } n \to \infty.$$

Then, any estimator $\mathcal{B}_n(\mathbf{Y}_n)$ satisfying

$$n^{\gamma^*}\{\mathcal{B}_n(\mathbf{Y}_n) - \beta_0\} = \Psi_n + o_{P_0}(1)$$

is regular and therefore attains the efficiency bound.

Proof Sketch

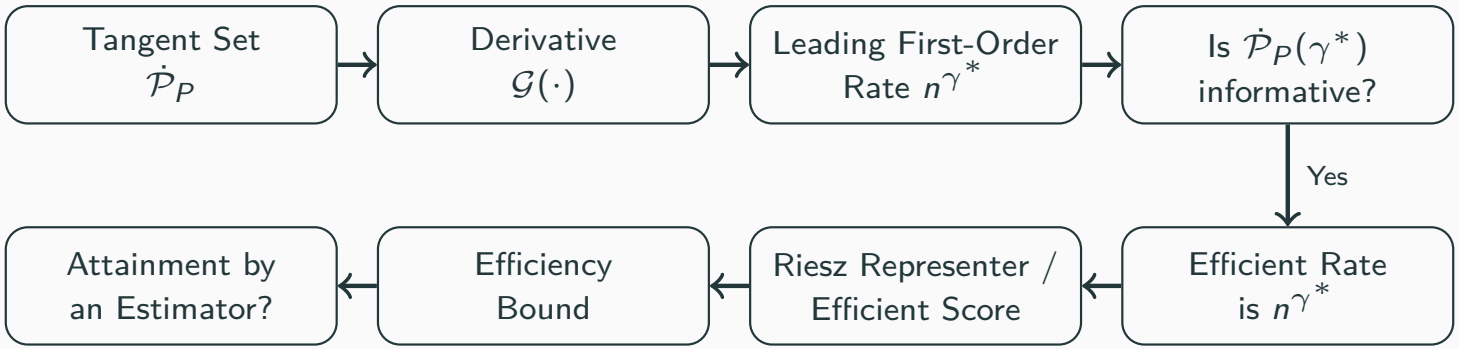
Running Example: The Simple Additive Model

- Joint normality holds trivially by the mutual independence of the errors ε_{ij} and the characteristics X_i
- For any $\gamma > \frac{1}{2}$,

$$n^{1/2-\gamma}\mathcal{G}(\{n^{\gamma}, S_n\}_{n\in N_+}) = \mathbb{E}_{P_0}\left[\sqrt{n}(\bar{Y}_n - \beta_0)S_n\right] = \mathbb{E}_{P_0}\left[\Psi_n S_n\right] + o(1)$$

  shows us that global efficiency must hold for Ψ_n
- Therefore, by theorem 3 any estimator that converges at rate $\sqrt{n}$ that is asymptotically linear in Ψ_n attains the efficiency bound—the sample mean in particular is efficient

Summary of Workflow



Contents

Introduction

Multi-Rate Semiparametric Efficiency Theory

Efficient Inference on the Mean in Network Models

- Efficiency Gains for the Simple Model
- A Non-Degenerate CID Model
- A Non-Degenerate CID Model with Additive Link Function

Conclusion

Appendix

- Proofs of Main Results
- Efficient Inference in Network Models with Exogenous Variables

Contents

Introduction

Multi-Rate Semiparametric Efficiency Theory

Efficient Inference on the Mean in Network Models

- Efficiency Gains for the Simple Model
  - A Non-Degenerate CID Model
  - A Non-Degenerate CID Model with Additive Link Function

Conclusion

Appendix

- Proofs of Main Results
- Efficient Inference in Network Models with Exogenous Variables

Efficiency Gains from Additional Model Restrictions

- Return to the simple model and suppose we know also that the mean of the characteristics X_i is equal to 0—this represents an additional model restriction

- The new model is described as

$$Y_{ij} = X_i + X_j + \varepsilon_{ij}, \quad \{\varepsilon_{ij}\}_{i,j\in\mathbb{N}_+} \perp \{X_i\}_{i\in\mathbb{N}_+}, \quad \mathbb{E}_P[X_i] = 0$$

- The tangent set $\dot{\mathcal{P}}_P$ is now the subset of the maximal tangent set where the pair $\{n^\gamma, S_n\}_{n\in N_+}$ satisfies

$$S_n = \frac{1}{n^\gamma}\left\{\sum_{i<j}^{n} q_\varepsilon(Y_{ij} - X_i - X_j) + \sum_{i=1}^{n} q_X(X_i)\right\}$$

for square integrable functions $q_\varepsilon(\cdot), q_X(\cdot)$ such that $\mathbb{E}_{P_0}\left[q_\varepsilon(\varepsilon_{ij})\right] = \mathbb{E}_{P_0}[q_X(X_i)] = 0$, **AND** $\mathbb{E}_{P_0}[X_i q_X(X_i)] = 0$

Efficiency Gains: $\mathcal{G}(\cdot)$

- For any one-dimensional submodel P_r with corresponding rate/score pair $\{n^\gamma, S_n\}_n$ that has edge-level score $q_\varepsilon(\cdot)$ and node-level score $q_X(\cdot)$,

$$\begin{aligned}
\frac{\partial}{\partial r}\beta(P_r)|_{r=0} &= \mathbb{E}_{P_0}\left[Y_{ij}\left\{q_\varepsilon(\varepsilon_{ij}) + q_X(X_i) + q_X(X_j)\right\}\right] \\
&= \mathbb{E}_{P_0}\left[(X_i + X_j + \varepsilon_{ij})\left\{q_\varepsilon(\varepsilon_{ij}) + q_X(X_i) + q_X(X_j)\right\}\right] \\
&= \mathbb{E}_{P_0}\left[\varepsilon_{ij} q_\varepsilon(\varepsilon_{ij})\right] + \mathbb{E}_{P_0}\left[X_i q_X(X_i)\right] + \mathbb{E}_{P_0}\left[X_j q_X(X_j)\right] \\
&= \mathbb{E}_{P_0}\left[\varepsilon_{ij} q_\varepsilon(\varepsilon_{ij})\right]
\end{aligned}$$

  so that the node-level scores $q_X(\cdot)$ do not enter into the derivative mapping at all

- Only tangent directions that admit edge-level perturbations have any first-order effects on the parameter of interest—what is the convergence rate of these directions?

Efficiency Gains: Leading First Order Rate

- Consider a rate/score pair $\{n^{\gamma}, S_n\}_{n \in N_+}$ with

$$S_n = \frac{1}{n^{\gamma}} \left\{ \sum_{i<j}^{n} q_{\varepsilon}(\varepsilon_{ij}) + \sum_{i=1}^{n} q_X(X_i) \right\}$$

- If $\gamma < 1$ and $\mathbb{E}_{P_0}\left[q_{\varepsilon}(\varepsilon_{ij})^2\right] > 0$ then

$$S_n = n^{1-\gamma} \times \underbrace{\frac{1}{n} \sum_{i<j}^{n} q_{\varepsilon}(\varepsilon_{ij})}_{O_p(1)} + n^{\frac{1}{2}-\gamma} \times \underbrace{\frac{1}{\sqrt{n}} \sum_{i=1}^{n} q_X(X_i)}_{O_p(1)}$$

  and $\{S_n\}_n$ is not tight—contradicts $S_n = O_p(1)$ and therefore we must have $q_{\varepsilon}(\varepsilon_{ij}) = 0$ almost surely

- If $\gamma > 1$ then $S_n \to 0$ which contradicts $\mathbb{I}(S) > 0$—$\dot{\mathcal{P}}(\gamma) = \emptyset$ for any $\gamma > 1$

- Implies that, as long as $\gamma < 1$, $V(\gamma) = 0$ and therefore that $\gamma^* = 1$

Efficiency Gains: Efficiency Bound

- Defining

$$\Psi_n = \frac{2}{n}\sum_{i<j}^{n}(\varepsilon_{ij} - \beta_0),$$

  $\{a\Psi_n\}_n \in \dot{\mathcal{P}}_P(1)$ by putting $q_\varepsilon(\varepsilon) = 2a(\varepsilon - \beta_0)$ and $q_X(\cdot) = 0$

- We also have

$$\mathcal{G}(\{n, S_n\}_n) = \lim_{n\to\infty} \mathbb{E}_{P_0}[\Psi_n S_n]$$

  so that the (global) efficiency condition is satisfied

- By theorem 1 n is the efficient rate of convergence and by theorem 2 the variance $2\sigma_\varepsilon^2$ of ε_{ij} is the efficiency bound

Efficiency Gains: Efficient Estimator

- Letting $\bar{X}_n$ be the sample average of the characteristics,

$$n\left\{\bar{Y}_n - 2\bar{X}_n - \beta_0\right\} = \Psi_n + o_{L^2}(1)$$

- By theorem 3 $\bar{Y}_n - 2\bar{X}_n$ is an estimator of the dyad mean that attains the efficiency bound

- Intuition: The restriction $\mathbb{E}_P[X_i] = 0$ represents an additional moment condition that we could exploit alongside $\mathbb{E}_P\left[Y_{ij} - \beta_0\right] = 0$—this additional information is used to deliver efficiency gains in the vein of Brown and Newey (1998)

- Novelty comes from the fact that exploiting the information in the additional moment actually gives rise to a faster regular estimator, and justifies the use of multi-rate framework because the rate of the obvious estimator (the sample mean) is not always the efficient rate

Contents

Introduction

Multi-Rate Semiparametric Efficiency Theory

Efficient Inference on the Mean in Network Models

- Efficiency Gains for the Simple Model
- A Non-Degenerate CID Model
- A Non-Degenerate CID Model with Additive Link Function

Conclusion

Appendix

- Proofs of Main Results
- Efficient Inference in Network Models with Exogenous Variables

Conditionally Independent Dyad Models

- Let $\{Y_{ij}\}_{i,j\in\mathbb{N}_+}$ be our data—for jointly exchangeable graphs/data, the Aldous-Hoover representation theorem furnishes us with a conditionally independent dyad (CID) model

$$Y_{ij} = g(\alpha, U_i, U_j, V_{ij}) \quad \text{for iid } \{U_i\}_{i\in\mathbb{N}_+},\ \{V_{ij}\}_{i,j\in\mathbb{N}_+}$$

  with a similar result furnished by Crane and Towsner (2018) when exogenous variables are present

- We focus attention on the model where observed dyadic data $\{Y_{ij}\}_{i<j,i,j\in\mathbb{N}_+}$ satisfies the CID restrictions

$$Y_{ij} \perp Y_{kl} \mid \{U_{i'}\}_{i'\in N_+}, \quad Y_{ij} \mid \{U_{i'}\}_{i'\in N_+} \sim Y_{ij} \mid U_i, U_j \tag{1}$$

  for some iid sequence of individual effects $\{U_i\}_{i\in N_+}$

Non-degeneracy in CID Models

- Parameter of interest is the dyad mean $\beta(P) = \mathbb{E}_P\left[Y_{ij}\right]$
- Letting P_0 be the true DGP, define $\beta_0 = \beta(P_0)$ and

$$h(u,u') = \mathbb{E}_{P_0}\left[Y_{ij} - \beta_0 \mid U_i = u, U_j = u'\right], \quad \bar{h}(u) = \mathbb{E}_{P_0}\left[h(u,U_j)\right]$$

  and assume non-degeneracy of the form $C_1 = \mathbb{E}_{P_0}\left[\bar{h}(U_i)^2\right] > 0$
- Non-degeneracy allows us to use Hajek projections to conclude that sample mean is asymptotically normal with $\sqrt{n}$ convergence rate:

$$\sqrt{n}(\bar{Y}_n - \beta_0) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} 2\bar{h}(U_i) + o_{L^2}(1) \overset{P_0}{\Rightarrow} \mathcal{N}\left[0, 4C_1\right]$$

Non-degenerate CID Models: The Tangent Set

Fisher's identity (prevalent in the state space literature, for example) shows us that the n-sample score is given as

$$\dot{\ell}_n(\mathbf{Y}_n) = \mathbb{E}_{P_0}\left[\dot{\ell}_n^{\mathcal{C}}(\mathbf{Y}_n, \mathbf{U}_n) \mid \mathbf{Y}_n\right]$$

and the joint score $\dot{\ell}_n^{\mathcal{C}}$ of the observed dyadic data and the individual effects can be decomposed as in the simple model

Definition 8 (Tangent Set for Non-degenerate CID Models)

Given the model restrictions in (1), the semiparametrically relevant tangent set $\dot{\mathcal{P}}_P$ consists of rate/score pairs $\{n^\gamma, S_n\}_{n \in N_+} \in \dot{\mathcal{P}}_{max}$ for which there exist square integrable functions $q_Y(\cdot), q_U(\cdot)$ such that

$$S_n = \frac{1}{n^\gamma}\left\{\sum_{i<j}^{n} \mathbb{E}_{P_0}\left[q_Y(Y_{ij}, U_i, U_j) \mid \mathbf{Y}_n\right] + \sum_{i=1}^{n} \mathbb{E}_{P_0}\left[q_U(U_i) \mid \mathbf{Y}_n\right]\right\}$$

and $\mathbb{E}_{P_0}\left[q_Y(Y_{ij}, U_i, U_j) \mid U_i, U_j\right] = \mathbb{E}_{P_0}\left[q_U(U_i)\right] = 0$.

Non-degenerate CID Models: The Mapping $\mathcal{G}(\cdot)$

- For any submodel P_r and corresponding rate/score pair $\{n^\gamma, S_n\}_{n \in N_+} \in \dot{\mathcal{P}}_P$,

$$\frac{\partial}{\partial r}\beta(P_r)|_{r=0} = n^\gamma \mathbb{E}_{P_0}\left[(\bar{Y}_n - \beta_0)S_n\right] \quad \forall n$$

- Non-degeneracy delivers $\sqrt{n}(\bar{Y}_n - \mu_0) = O_{P_0}(1)$, so

$$\mathcal{G}(\{n^\gamma, S_n\}_n) = n^{\gamma - \frac{1}{2}} \underbrace{\mathbb{E}_{P_0}\left[\sqrt{n}(\bar{Y}_n - \mu_0)S_n\right]}_{O(1)}$$

- We can immediately tell that any rate n^γ such that $\gamma < \frac{1}{2}$ (slower than $\sqrt{n}$) has a first-order noninformative stratum

Non-degenerate CID Models: Is the $n^{1/2}$ stratum informative?

- If $\dot{\mathcal{P}}_P(1/2)$ is first-order informative, then $n^{1/2}$ is the leading first-order rate as well as the efficient rate
- Define

$$\Psi_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \mathbb{E}_{P_0} \left[2\bar{h}(U_i) \mid \mathbf{Y}_n \right],$$

which is asymptotically equivalent to $\sqrt{n}(\bar{Y}_n - \mu_0)$ and therefore satisfies $\Psi_n \overset{P_0}{\Rightarrow} \mathcal{N}[0, 4C_1]$ and $\mathbb{E}_{P_0}\left[\Psi_n^2\right] \to 4C_1$

Lemma 2 (Tangency and Efficiency of Ψ_n)

$\{n^{1/2}, a\Psi_n\}_{n \in N_+} \in \dot{\mathcal{P}}_P$ for any non-zero $a \in \mathbb{R}$ and

$$\left| \mathbb{E}_{P_0}[\Psi_n S_n] - n^{\frac{1}{2}-\gamma} \mathcal{G}(\{n^\gamma, S_n\}_{n \in N_+}) \right| = o(1)$$

for any $\{n^\gamma, S_n\}_n \in \dot{\mathcal{P}}_P$.

Non-degenerate CID Models: The Efficiency Bound and Attainment

- Implication of global efficiency condition is that $\{\Psi_n\}_{n\in N_+} \in \dot{\mathcal{P}}_P(1/2)$ has derivative equal to $4C_1 > 0$, so that $\dot{\mathcal{P}}_P(1/2)$ is first-order informative—$n^{1/2}$ is the leading first order rate as well as the efficient rate of convergence

- $\{\Psi_n\}_{n\in N_+}$ satisfies conditions of theorem 2, and therefore $4C_1$ is the efficiency bound

- Global efficiency condition in theorem 3 also holds: so the sample mean $\bar{Y}_n$ attains the efficiency bound as long as the joint normality holds, with the condition below being sufficient

Joint Convergence Condition *For any finite collection of score sequences* $\{\{S_{i,n}\}_{n\in N_+}\}_{1\leq i\leq t}$ *that belong to rate/score pairs in* $\dot{\mathcal{P}}_P$*, the* t*-dimensional random vector* $(S_{1,n}, \cdots, S_{t,n})$ *converges jointly to a normal distribution under* P_0

Contents

Introduction

Multi-Rate Semiparametric Efficiency Theory

Efficient Inference on the Mean in Network Models

- Efficiency Gains for the Simple Model
- A Non-Degenerate CID Model
- A Non-Degenerate CID Model with Additive Link Function

Conclusion

Appendix

- Proofs of Main Results
- Efficient Inference in Network Models with Exogenous Variables

Additive CID Models

- We go one step further and consider the efficiency bound in the model (1) with an additive graphon structure
- The new model restrictions are

$$Y_{ij} = U_i + U_j + V_{ij}, \quad \{U_i\}_{i \in N_+} \perp \{V_{jk}\}_{j,k \in N_+, j<k} \tag{2}$$

- The tangent space compared to that of the model (1) only requires an additional functional form restriction on the edge-level scores, which can be expressed in terms of the score for V_{ij} under additivity
- Non-degeneracy assumption + parameter of interest + derivative mapping continue to remain unchanged, and therefore sample mean still converges at rate $\sqrt{n}$

Additive CID Models: Tangent Set

Score sequences for the additive CID model requires conditioning on observed dyads as in the unrestricted CID case + involves the V_{ij}-score as in the simple additive model with observed characteristics

Definition 9 (Tangent Set for Additive CID Model)

Given the model restrictions in (2), the semiparametrically relevant tangent set $\dot{\mathcal{P}}_+$ consists of rate/score pairs $\{n^\gamma, S_n\}_{n \in N_+} \in \dot{\mathcal{P}}_P$ for which there exist square integrable functions $q_V(\cdot), q_U(\cdot)$ such that

$$S_n = \frac{1}{n^\gamma} \left\{ \sum_{i<j}^{n} \mathbb{E}_{P_0} \left[q_V(Y_{ij} - U_i - U_j) \mid \mathbf{Y}_n \right] + \sum_{i=1}^{n} \mathbb{E}_{P_0} \left[q_U(U_i) \mid \mathbf{Y}_n \right] \right\}$$

and $\mathbb{E}_{P_0} \left[q_V(V_{ij}) \mid U_i, U_j \right] = \mathbb{E}_{P_0} \left[q_U(U_i) \right] = 0.$

Additive CID Models: Adaptivity of the Sample Mean

- Recall that the efficient score for the unrestricted CID model involved only node-level perturbations—the additional restrictions imposed by the additive link function affects only the edge-level score
- As such, $\Psi_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \mathbb{E}_{P_0} [2U_i \mid \mathbf{Y}_n]$ continues to be the efficient score, lemma 2 continues to hold, and the sample mean continues to be efficient under joint normality constraint
- Knowledge of additive link function does not bring efficiency gains—the sample mean is adaptive
- Going further: knowledge of any (not necessarily additive) link function only affects the edge-level score and therefore the sample mean remains adaptive

Contents

Introduction

Multi-Rate Semiparametric Efficiency Theory

Efficient Inference on the Mean in Network Models

- Efficiency Gains for the Simple Model
- A Non-Degenerate CID Model
- A Non-Degenerate CID Model with Additive Link Function

Conclusion

Appendix

- Proofs of Main Results
- Efficient Inference in Network Models with Exogenous Variables

Conclusion

- Generally difficult to establish semiparametric efficiency in models with dyadic data because generically, different tangent directions converge at different rates
- Develop new multi-rate semiparametric efficiency theory—categorize tangent directions into different strata, find efficient rate, and then search for efficiency bound within the efficient stratum
- New theorems for finding the efficient rate; establishing efficiency bound via the convolution theorem; and for concluding that an estimator attains the bound
- Inference on the dyadic mean in an additive model with observed characteristics reveals efficient rate and estimator depends on the kinds of restrictions imposed
- From application to inference on the dyadic mean in a CID model we learn that the sample mean is efficient and adaptive to additional knowledge of the link function

Contents

Introduction

Multi-Rate Semiparametric Efficiency Theory

Efficient Inference on the Mean in Network Models

- Efficiency Gains for the Simple Model
- A Non-Degenerate CID Model
- A Non-Degenerate CID Model with Additive Link Function

Conclusion

Appendix

- Proofs of Main Results
- Efficient Inference in Network Models with Exogenous Variables

Contents

Introduction

Multi-Rate Semiparametric Efficiency Theory

Efficient Inference on the Mean in Network Models

- Efficiency Gains for the Simple Model
- A Non-Degenerate CID Model
- A Non-Degenerate CID Model with Additive Link Function

Conclusion

Appendix

- Proofs of Main Results
- Efficient Inference in Network Models with Exogenous Variables

Proof Sketch of Lemma 1

Back

For the sake of contradiction suppose that $\dot{\mathcal{P}}_P(\gamma^*)$ is first-order informative, so that there exists a submodel P_r with score sequence $\{S_n\}_n \in \dot{\mathcal{P}}_P(\gamma^*)$ such that $\mathcal{G}(\{n^{\gamma^*}, S_n\}_n) \neq O_{k\times 1}$. Let $n^\kappa \{\mathcal{B}_n(\mathbf{Y}_n) - \beta_0\}$ converge to some distribution $\mathcal{L}$ under P_0.

Since a LAN expansion for $\log \frac{dP_{n,n^{-\gamma^*}}}{dP_{n,0}}(\mathbf{Y}_n)$ holds by definition of the tangent set, Le Cam's third lemma implies that $n^\kappa \{\mathcal{B}_n(\mathbf{Y}_n) - \beta_0\}$ converges to some distribution $\mathcal{L}^*$ under $P_{n^{-\gamma^*}}$. By regularity,

$$n^\kappa \{\mathcal{B}_n(\mathbf{Y}_n) - \beta(P_{n^{-\gamma^*}})\} = n^\kappa \{\mathcal{B}_n(\mathbf{Y}_n) - \beta_0\} - n^{\kappa-\gamma^*} \frac{\beta(P_{n^{-\gamma^*}}) - \beta(P_0)}{n^{-\gamma^*}}$$

converges to $\mathcal{L}$ under $P_{n^{-\gamma^*}}$. $\kappa > \gamma^*$, so the second term diverges to $+\infty$, meaning that $n^\kappa \{\mathcal{B}_n(\mathbf{Y}_n) - \beta(P_{n^{-\gamma^*}})\}$, being the sum of a tight element and a divergent element, is not tight under $P_{n^{-\gamma^*}}$. This contradicts the fact that this quantity converges to a probability distribution under $P_{n^{-\gamma^*}}$.

Proof Sketch of Theorem 2

Back

By assumption

$$n^{\gamma^*}\left\{\mathcal{B}_n(\mathbf{Y}_n) - \beta_0\right\} \overset{P_{n-\gamma^*}}{\Rightarrow} \mathcal{L}$$

for any submodel P_r with score sequence in the efficient stratum $\{S_n\}_n \in \dot{\mathcal{P}}_P(\gamma^*)$—this includes $\{h'\Psi_n\}_n$ for any non-zero $h \in \mathbb{R}^k$ by tangency. Focus our attention on the k-dimensional parametric submodel with score Ψ_n; because $\mathcal{B}_n(\mathbf{Y}_n)$ is a regular estimator in the usual sense for this parametric model, the parametric convolution theorem tells us that

$$\mathcal{L} = \mathcal{N}\left[O_{k\times 1},\ \dot{\beta}_0 V_0^{-1} \dot{\beta}_0'\right]$$

where

$$\dot{\beta}_0 = \begin{pmatrix} \mathcal{G}(\{n^{\gamma^*}, e_1'\Psi_n\}_n)' \\ \vdots \\ \mathcal{G}(\{n^{\gamma^*}, e_k'\Psi_n\}_n)' \end{pmatrix}, \quad \{e_1, \cdots, e_k\} \subset \mathbb{R}^k \text{ is the standard basis}$$

By local efficiency, $\dot{\beta}_0 = V_0$ and we have the desired result.

Proof Sketch of Theorem 3

Back

We need to prove regularity of the estimator $\mathcal{B}_n(\mathbf{Y}_n)$. For any $\{n^\gamma, S_n\}_n \in \dot{\mathcal{P}}_P$ with $\gamma \geq \gamma^*$, $\log \frac{dP_{n,n^{-\gamma}}}{dP_{n,0}}(\mathbf{Y}_n)$ is asymptotically linear in $S_n - \frac{1}{2}\mathbb{I}(S)$, while $\mathcal{B}_n(\mathbf{Y}_n)$ is asymptotically linear in Ψ_n—by joint normality and Le Cam's third lemma,

$$n^{\gamma^*}\{\mathcal{B}_n(\mathbf{Y}_n) - \beta_0\} \overset{P_{n^{-\gamma}}}{\Rightarrow} \mathcal{N}[\tau, V_0]$$

for location shift $\tau = \lim_{n\to\infty} \mathbb{E}_{P_0}[\Psi_n S_n]$. Use the decomposition

$$n^{\gamma^*}\{\mathcal{B}_n(\mathbf{Y}_n) - \beta(P_{n^{-\gamma}})\} = n^{\gamma^*}\{\mathcal{B}_n(\mathbf{Y}_n) - \beta_0\} - n^{\gamma^*-\gamma}\frac{\beta(\beta(P_{n^{-\gamma}}) - \beta(P_0)}{n^{-\gamma}}.$$

If $\gamma^* = \gamma$, then the second term converges to $\mathcal{G}(\{n^\gamma, S_n\}_n)$ as $n \to \infty$. If $\gamma^* < \gamma$, then the second term converges to 0. In any case, the second term converges to τ by global efficiency. Therefore, by Slutsky's theorem

$$n^{\gamma^*}\{\mathcal{B}_n(\mathbf{Y}_n) - \beta(P_{n^{-\gamma}})\} \overset{P_{n^{-\gamma}}}{\Rightarrow} \mathcal{N}[\tau, V_0] - \tau \sim \mathcal{N}[O_{k\times 1}, V_0]$$

and $\mathcal{B}_n(\mathbf{Y}_n)$ is regular.

Contents

Introduction

Multi-Rate Semiparametric Efficiency Theory

Efficient Inference on the Mean in Network Models

- Efficiency Gains for the Simple Model
- A Non-Degenerate CID Model
- A Non-Degenerate CID Model with Additive Link Function

Conclusion

Appendix

- Proofs of Main Results
- Efficient Inference in Network Models with Exogenous Variables

CID Models with Exogenous Variables

- Straightforward generalization of the mean inference case when there are exogenous unit-specific characteristics $\{X_i\}_{i \in N_+}$ alongside dyadic data $\{Y_{ij}\}_{i,j \in N_+, i<j}$
- Assume CID models conditional on the exogenous variables:

$$Y_{ij} \perp Y_{kl} \mid \mathbf{U}, \mathbf{X}, \quad Y_{ij} \mid \mathbf{U}, \mathbf{X} \sim Y_{ij} \mid X_i, U_i, X_j, U_j \tag{3}$$

  where $\{U_i\}_{i \in N_+}$ are iid individual effects such that $\{Z_i = (X_i, U_i)\}_{i \in N_+}$ is also iid

- Let P_0 be the true DGP and suppose the parameter mapping of interest is

$$\beta(P) = g(\mathbb{E}_P \left[\phi(Y_{ij}, X_i, X_j)\right])$$

  for some vector valued function $\phi(\cdot)$ and transformation $g(\cdot)$—denote $\alpha(P) = \mathbb{E}_P \left[\phi(Y_{ij}, X_i, X_j)\right]$

Exogenous Variables: The Tangent Set

Definition 10 (Tangent Set for CID Models with Exogenous Variables)

Given the model restrictions in (3), the semiparametrically relevant tangent set $\dot{\mathcal{P}}_P$ consists of rate/score pairs $\{n^\gamma, S_n\}_{n \in N_+} \in \dot{\mathcal{P}}_{max}$ for which there exist square integrable functions $q_Y(\cdot), q_Z(\cdot)$ such that

$$S_n = \frac{1}{n^\gamma} \left\{ \sum_{i<j}^{n} \mathbb{E}_{P_0} \left[q_Y(Y_{ij}, Z_i, Z_j) \mid \mathbf{Y}_n \right] + \sum_{i=1}^{n} \mathbb{E}_{P_0} \left[q_Z(Z_i) \mid \mathbf{Y}_n \right] \right\}$$

and $\mathbb{E}_{P_0} \left[q_Y(Y_{ij}, Z_i, Z_j) \mid Z_i, Z_j \right] = \mathbb{E}_{P_0} \left[q_Z(Z_i) \right] = 0.$

Exogenous Variables: Non-degeneracy

- Define

$$h(u,x,u',x') = \mathbb{E}_{P_0}\left[\phi(Y_{ij},X_i,X_j) - \alpha_0 \mid U_i = u, X_i = x, U_j = u', X_j = x'\right]$$

$$\bar{h}(u,x) = \mathbb{E}_{P_0}\left[h(u,x,U_j,X_j)\right]$$

- We say that the model is non-degenerate if $\Gamma_0 = \mathbb{E}_{P_0}\left[\bar{h}(U_i,X_i)\bar{h}(U_i,X_i)'\right]$ is positive definite

- Letting

$$\bar{\mathcal{A}}_n = \frac{1}{\binom{n}{2}} \sum_{i<j}^{n} \phi(Y_{ij},X_i,X_j),$$

under non-degeneracy the same Hajek projection construction as before reveals that

$$\sqrt{n}\left\{\bar{\mathcal{A}}_n - \alpha_0\right\} \overset{P_0}{\Rightarrow} \mathcal{N}\left[O_{m\times 1},\ 4\Gamma_0\right]$$

Exogenous Variables: Derivative Mapping

- If $g(\cdot)$ is differentiable at $\mathbb{E}_{P_0}\left[\phi(Y_{ij}, X_i, X_j)\right]$ then, for any $\{n^\gamma, S_n\}_{n \in N_+} \in \dot{\mathcal{P}}_P$,

$$\mathcal{G}(\{n^\gamma, S_n\}_{n \in N_+}) = \frac{\partial}{\partial r}\beta(P_r)|_{r=0}$$

$$= \underbrace{g'(\mathbb{E}_{P_0}\left[\phi(Y_{ij}, X_i, X_j)\right])}_{Q} \times n^{\gamma - \frac{1}{2}} \underbrace{\mathbb{E}_{P_0}\left[\sqrt{n}(\bar{\mathcal{A}}_n - \alpha_0) S_n\right]}_{O(1)}$$

- Once again we can see that $\mathcal{G}(\{n^\gamma, S_n\}_{n \in N_+}) = O_{k \times 1}$ for any $\gamma < \frac{1}{2}$, so that $\gamma^* \geq \frac{1}{2}$

Exogenous Variables: Efficiency Bound and Attainment

- Defining

$$\Psi_n = Q\frac{1}{\sqrt{n}}\sum_{i=1}^{n} 2\bar{h}(U_i, X_i)$$

  we know that $\sqrt{n}\left\{Q\bar{\mathcal{A}}_n - \beta_0\right\} = \Psi_n + o_{L^2}(1)$

- The same process as in the mean inference case shows us that $\{\Psi_n\}_{n\in N_+}$ satisfies the conditions of theorem 2—by implication, $n^{1/2}$ is the leading first order/efficient rate and $4Q\Gamma_0 Q'$ is the efficiency bound

- To prove that $g(\mathcal{A}_n)$ attains the efficiency bound we additionally require the joint convergence assumption as before; global efficiency follows from the non-degeneracy assumption